In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

```
C:\Users\Shaki\anaconda3\Lib\site-packages\pandas\core\arrays\masked.py:60: U
serWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version
'1.3.5' currently installed).
  from pandas.core import (
```

In [2]:
```
# Load dataset
df = pd.read_csv('household_power_consumption.txt',
                 sep=';',
                 na_values=['?'],
                 low_memory=False)
```

In [3]:
```
# Basic exploration
print("Shape:", df.shape)
print("\nDtypes:\n", df.dtypes)
print("\nMissing values:\n", df.isnull().sum())
print("\nBasic Stats:\n", df.describe())
print("\nFirst 5 rows:\n", df.head())
```

```
Shape: (2075259, 9)

Dtypes:
 Date                     object
Time                     object
Global_active_power     float64
Global_reactive_power   float64
Voltage                 float64
Global_intensity        float64
Sub_metering_1          float64
Sub_metering_2          float64
Sub_metering_3          float64
dtype: object

Missing values:
 Date                        0
Time                        0
Global_active_power     25979
Global_reactive_power   25979
Voltage                 25979
Global_intensity        25979
Sub_metering_1          25979
Sub_metering_2          25979
Sub_metering_3          25979
dtype: int64

Basic Stats:
        Global_active_power  Global_reactive_power       Voltage  \
count          2.049280e+06           2.049280e+06  2.049280e+06
mean           1.091615e+00           1.237145e-01  2.408399e+02
std            1.057294e+00           1.127220e-01  3.239987e+00
min            7.600000e-02           0.000000e+00  2.232000e+02
25%            3.080000e-01           4.800000e-02  2.389900e+02
50%            6.020000e-01           1.000000e-01  2.410100e+02
75%            1.528000e+00           1.940000e-01  2.428900e+02
max            1.112200e+01           1.390000e+00  2.541500e+02

       Global_intensity  Sub_metering_1  Sub_metering_2  Sub_metering_3
count      2.049280e+06    2.049280e+06    2.049280e+06    2.049280e+06
mean       4.627759e+00    1.121923e+00    1.298520e+00    6.458447e+00
std        4.444396e+00    6.153031e+00    5.822026e+00    8.437154e+00
min        2.000000e-01    0.000000e+00    0.000000e+00    0.000000e+00
25%        1.400000e+00    0.000000e+00    0.000000e+00    0.000000e+00
50%        2.600000e+00    0.000000e+00    0.000000e+00    1.000000e+00
75%        6.400000e+00    0.000000e+00    1.000000e+00    1.700000e+01
max        4.840000e+01    8.800000e+01    8.000000e+01    3.100000e+01

First 5 rows:
         Date      Time  Global_active_power  Global_reactive_power  Voltage
\
0  16/12/2006  17:24:00                4.216                  0.418   234.84
1  16/12/2006  17:25:00                5.360                  0.436   233.63
2  16/12/2006  17:26:00                5.374                  0.498   233.29
3  16/12/2006  17:27:00                5.388                  0.502   233.74
4  16/12/2006  17:28:00                3.666                  0.528   235.68

   Global_intensity  Sub_metering_1  Sub_metering_2  Sub_metering_3
```

0	18.4	0.0	1.0	17.0
1	23.0	0.0	1.0	16.0
2	23.0	0.0	2.0	17.0
3	23.0	0.0	1.0	17.0
4	15.8	0.0	1.0	17.0

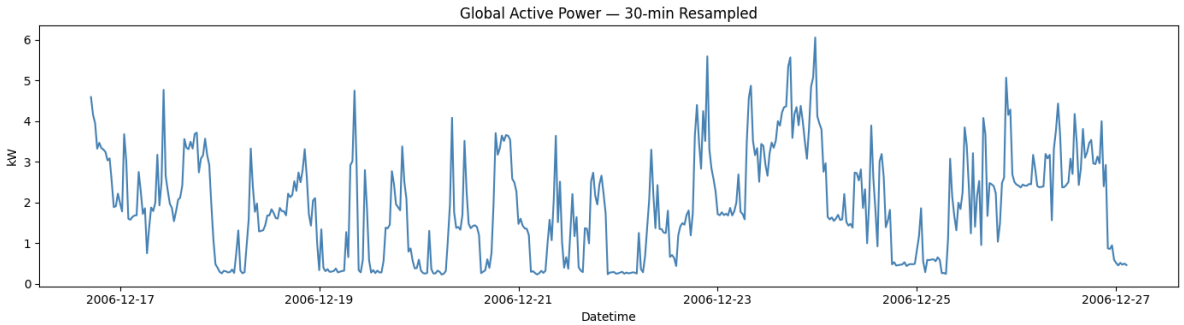
In [14]:
```
# Work on 30-min resampled data
df_feat = df_30min.copy()

# Datetime features
df_feat['hour']        = df_feat.index.hour
df_feat['dayofweek']   = df_feat.index.dayofweek
df_feat['month']       = df_feat.index.month
df_feat['quarter']     = df_feat.index.quarter
df_feat['is_weekend']  = (df_feat.index.dayofweek >= 5).astype(int)

# Lag features
df_feat['lag_1']  = df_feat['Global_active_power'].shift(1)
df_feat['lag_2']  = df_feat['Global_active_power'].shift(2)

# Rolling mean — FIXED (.mean() added)
df_feat['rolling_mean_4'] = df_feat['Global_active_power'].shift(1).rolling(4)

# Drop NaN rows
df_feat.dropna(inplace=True)

# Final 9 features — FIXED (Global_intensity not Global_voltage)
features = ['hour', 'dayofweek', 'month', 'quarter', 'is_weekend',
            'lag_1', 'lag_2', 'rolling_mean_4',
            'Global_intensity']

print("Feature columns:", features)
print("Shape:", df_feat.shape)
```

```
Feature columns: ['hour', 'dayofweek', 'month', 'quarter', 'is_weekend', 'lag
_1', 'lag_2', 'rolling_mean_4', 'Global_intensity']
Shape: (69173, 15)
```

In [ ]:

In [15]:
```
# Resample to 30-minute frequency
df_30min = df.resample('30T').mean()

print("Shape after resampling:", df_30min.shape)
print(df_30min.head())

# Plot to verify smoothing
plt.figure(figsize=(14, 4))
plt.plot(df_30min['Global_active_power'][:500], color='steelblue')
plt.title('Global Active Power — 30-min Resampled')
plt.xlabel('Datetime')
plt.ylabel('kW')
plt.tight_layout()
plt.show()
```

```
Shape after resampling: (69177, 7)
                     Global_active_power  Global_reactive_power     Voltage
\
Datetime
2006-12-16 17:00:00             4.587333               0.484000  234.366667
2006-12-16 17:30:00             4.150000               0.178000  234.699333
2006-12-16 18:00:00             3.944800               0.084867  235.184667
2006-12-16 18:30:00             3.319600               0.075200  233.975667
2006-12-16 19:00:00             3.464400               0.049800  233.754000

                     Global_intensity  Sub_metering_1  Sub_metering_2  \
Datetime
2006-12-16 17:00:00         19.700000             0.0        1.333333
2006-12-16 17:30:00         17.780000             0.0        0.366667
2006-12-16 18:00:00         16.966667             0.0       11.400000
2006-12-16 18:30:00         14.233333             0.0        2.033333
2006-12-16 19:00:00         14.746667             0.0        1.833333

                     Sub_metering_3
Datetime
2006-12-16 17:00:00       16.833333
2006-12-16 17:30:00       16.866667
2006-12-16 18:00:00       16.966667
2006-12-16 18:30:00       16.766667
2006-12-16 19:00:00       16.766667
```



In [20]:
```
df_feat = df_30min.copy()

df_feat['hour']        = df_feat.index.hour
df_feat['dayofweek']   = df_feat.index.dayofweek
df_feat['month']       = df_feat.index.month
df_feat['quarter']     = df_feat.index.quarter
df_feat['is_weekend']  = (df_feat.index.dayofweek >= 5).astype(int)

df_feat['lag_1']  = df_feat['Global_active_power'].shift(1)
df_feat['lag_2']  = df_feat['Global_active_power'].shift(2)

df_feat['rolling_mean_4'] = df_feat['Global_active_power'].shift(1).rolling(4)

df_feat.dropna(inplace=True)

features = ['hour', 'dayofweek', 'month', 'quarter', 'is_weekend',
            'lag_1', 'lag_2', 'rolling_mean_4',
            'Global_intensity']

print(features)
print(df_feat.shape)
```

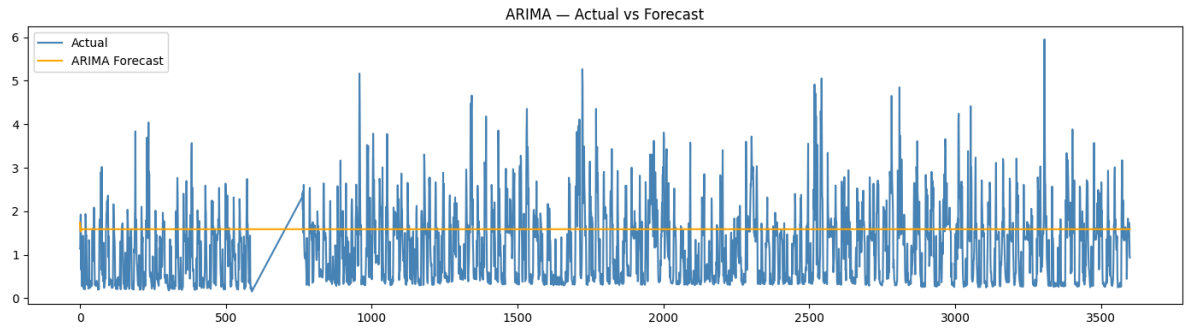
```
['hour', 'dayofweek', 'month', 'quarter', 'is_weekend', 'lag_1', 'lag_2', 'ro
lling_mean_4', 'Global_intensity']
(69173, 15)
```

In [21]:
```
print(features)
```

```
['hour', 'dayofweek', 'month', 'quarter', 'is_weekend', 'lag_1', 'lag_2', 'ro
lling_mean_4', 'Global_intensity']
```

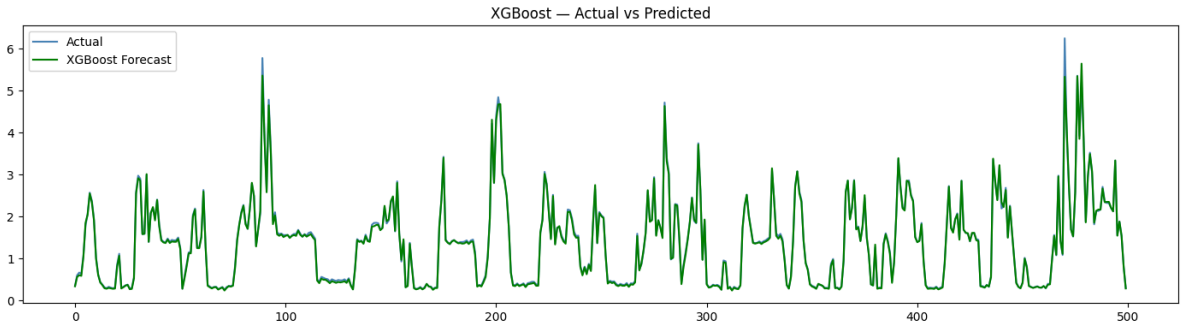
In [22]:
```
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error

# Use last 18K records of 30-min resampled data
series = df_30min['Global_active_power'].dropna()
series = series[-18000:]

# Train/test split — 80/20
split = int(len(series) * 0.8)
train_arima = series[:split]
test_arima  = series[split:]

print(f"ARIMA Train: {len(train_arima)} | Test: {len(test_arima)}")

# Fit ARIMA(5,1,0)
model_arima = ARIMA(train_arima, order=(5, 1, 0))
model_arima_fit = model_arima.fit()

# Forecast
forecast_arima = model_arima_fit.forecast(steps=len(test_arima))

# RMSE
rmse_arima = np.sqrt(mean_squared_error(test_arima, forecast_arima))
print(f"ARIMA RMSE: {rmse_arima:.4f}")

# Plot
plt.figure(figsize=(14, 4))
plt.plot(test_arima.values, label='Actual', color='steelblue')
plt.plot(forecast_arima.values, label='ARIMA Forecast', color='orange')
plt.title('ARIMA — Actual vs Forecast')
plt.legend()
plt.tight_layout()
plt.show()
```

```
ARIMA Train: 14400 | Test: 3600
ARIMA RMSE: 0.9616
```



In [23]:
```
from xgboost import XGBRegressor
from sklearn.model_selection import GridSearchCV, train_test_split
from sklearn.metrics import mean_squared_error

# Features and target
X = df_feat[features]
y = df_feat['Global_active_power']

# Train/test split — no shuffle (time-series order matters)
split = int(len(X) * 0.8)
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

print(f"XGBoost Train: {len(X_train)} | Test: {len(X_test)}")

# Grid search
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [3, 5],
    'learning_rate': [0.05, 0.1],
    'subsample': [0.8, 1.0]
}

xgb = XGBRegressor(random_state=42, verbosity=0)
grid_search = GridSearchCV(xgb, param_grid, cv=3,
                           scoring='neg_root_mean_squared_error',
                           n_jobs=-1, verbose=1)
grid_search.fit(X_train, y_train)

print("Best params:", grid_search.best_params_)

# Best model
best_xgb = grid_search.best_estimator_
y_pred_xgb = best_xgb.predict(X_test)

# RMSE
rmse_xgb = np.sqrt(mean_squared_error(y_test, y_pred_xgb))
print(f"XGBoost RMSE: {rmse_xgb:.4f}")

# Plot
plt.figure(figsize=(14, 4))
plt.plot(y_test.values[:500], label='Actual', color='steelblue')
plt.plot(y_pred_xgb[:500], label='XGBoost Forecast', color='green')
plt.title('XGBoost — Actual vs Predicted')
plt.legend()
plt.tight_layout()
plt.show()
```

```
XGBoost Train: 55338 | Test: 13835
Fitting 3 folds for each of 16 candidates, totalling 48 fits
Best params: {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200, 'su
bsample': 0.8}
XGBoost RMSE: 0.0340
```

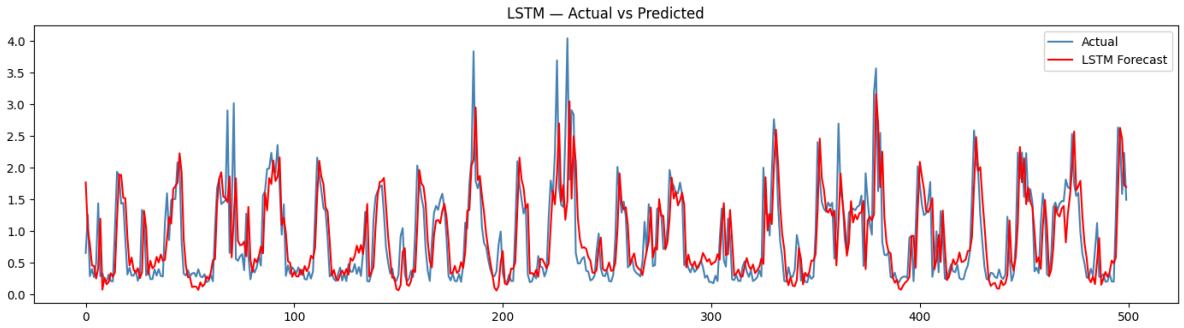

XGBoost — Actual vs Predicted
Actual
XGBoost Forecast
0
1
2
3
4
5
6
0
100
200
300
400
500

In [24]:

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error

# 4 input features for multivariate LSTM
lstm_features = ['Global_active_power', 'Sub_metering_1',
                 'Sub_metering_2', 'Sub_metering_3']

lstm_data = df_30min[lstm_features].dropna()
lstm_data = lstm_data[-18000:]  # Latest 18K records

# Scale
scaler = MinMaxScaler()
scaled = scaler.fit_transform(lstm_data)

# Create sequences — 24 timesteps to predict next value
def create_sequences(data, seq_len=24):
    X, y = [], []
    for i in range(len(data) - seq_len):
        X.append(data[i:i+seq_len])          # all 4 features
        y.append(data[i+seq_len][0])          # only Global_active_power
    return np.array(X), np.array(y)

SEQ_LEN = 24
X_lstm, y_lstm = create_sequences(scaled, SEQ_LEN)

# Train/test split
split = int(len(X_lstm) * 0.8)
X_train_l, X_test_l = X_lstm[:split], X_lstm[split:]
y_train_l, y_test_l = y_lstm[:split], y_lstm[split:]

print(f"LSTM Train: {X_train_l.shape} | Test: {X_test_l.shape}")

# Build LSTM model
model_lstm = Sequential([
    LSTM(64, return_sequences=True, input_shape=(SEQ_LEN, 4)),
    Dropout(0.2),
    LSTM(32),
    Dropout(0.2),
    Dense(1)
])

model_lstm.compile(optimizer='adam', loss='mse')
model_lstm.summary()

# Train
history = model_lstm.fit(X_train_l, y_train_l,
                         epochs=10,
                         batch_size=64,
                         validation_split=0.1,
                         verbose=1)

# Predict
y_pred_lstm_scaled = model_lstm.predict(X_test_l)
```

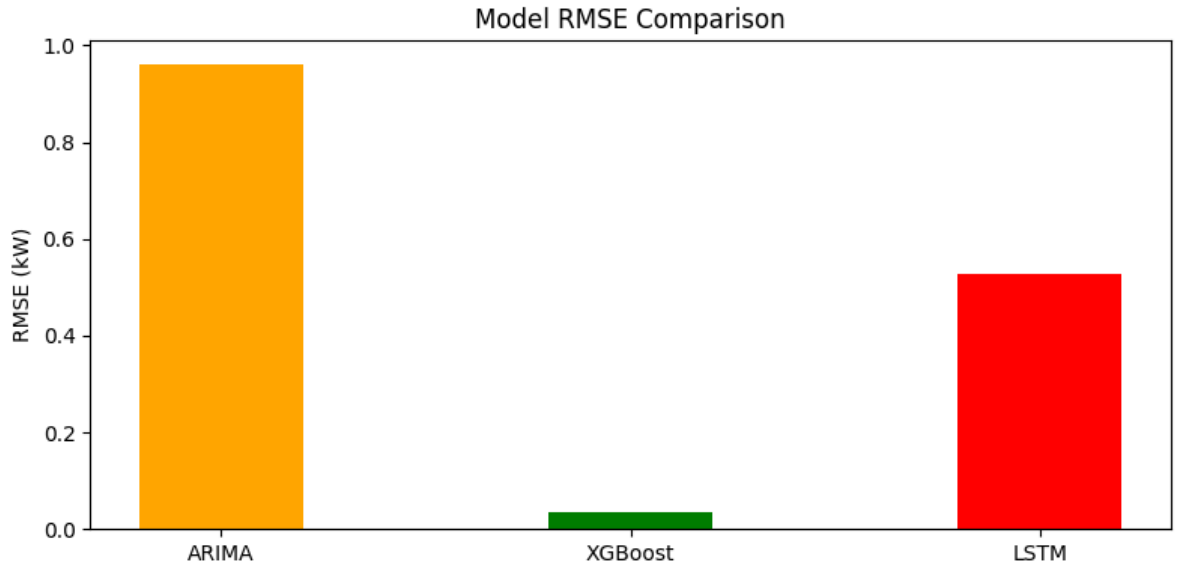
```python
# Inverse transform — only Global_active_power column
def inverse_target(scaled_vals, scaler, n_features=4):
    dummy = np.zeros((len(scaled_vals), n_features))
    dummy[:, 0] = scaled_vals.flatten()
    return scaler.inverse_transform(dummy)[:, 0]

y_pred_lstm = inverse_target(y_pred_lstm_scaled, scaler)
y_test_lstm  = inverse_target(y_test_l, scaler)

# RMSE
rmse_lstm = np.sqrt(mean_squared_error(y_test_lstm, y_pred_lstm))
print(f"LSTM RMSE: {rmse_lstm:.4f}")

# Plot
plt.figure(figsize=(14, 4))
plt.plot(y_test_lstm[:500], label='Actual', color='steelblue')
plt.plot(y_pred_lstm[:500], label='LSTM Forecast', color='red')
plt.title('LSTM — Actual vs Predicted')
plt.legend()
plt.tight_layout()
plt.show()
```

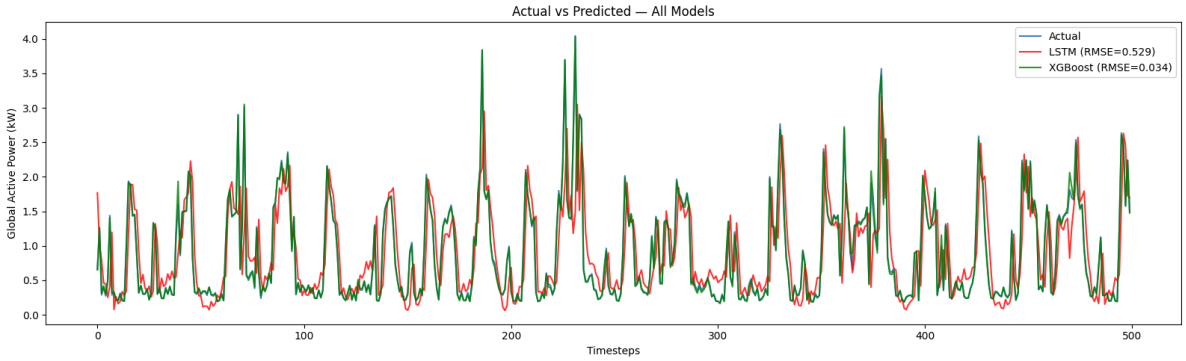
```
LSTM Train: (14380, 24, 4) | Test: (3596, 24, 4)
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 lstm_2 (LSTM)               (None, 24, 64)            17664

 dropout_2 (Dropout)         (None, 24, 64)            0

 lstm_3 (LSTM)               (None, 32)                12416

 dropout_3 (Dropout)         (None, 32)                0

 dense_1 (Dense)             (None, 1)                 33

=================================================================
Total params: 30113 (117.63 KB)
Trainable params: 30113 (117.63 KB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
Epoch 1/10
203/203 [==============================] - 25s 73ms/step - loss: 0.0135 - val
_loss: 0.0061
Epoch 2/10
203/203 [==============================] - 11s 53ms/step - loss: 0.0099 - val
_loss: 0.0052
Epoch 3/10
203/203 [==============================] - 11s 53ms/step - loss: 0.0087 - val
_loss: 0.0049
Epoch 4/10
203/203 [==============================] - 10s 47ms/step - loss: 0.0080 - val
_loss: 0.0045
Epoch 5/10
203/203 [==============================] - 8s 39ms/step - loss: 0.0078 - val_
loss: 0.0044
Epoch 6/10
203/203 [==============================] - 8s 39ms/step - loss: 0.0076 - val_
loss: 0.0044
Epoch 7/10
203/203 [==============================] - 9s 44ms/step - loss: 0.0074 - val_
loss: 0.0046
Epoch 8/10
203/203 [==============================] - 11s 52ms/step - loss: 0.0073 - val
_loss: 0.0044
Epoch 9/10
203/203 [==============================] - 9s 46ms/step - loss: 0.0072 - val_
loss: 0.0046
Epoch 10/10
203/203 [==============================] - 11s 55ms/step - loss: 0.0071 - val
_loss: 0.0043
113/113 [==============================] - 4s 14ms/step
LSTM RMSE: 0.5287
```


LSTM — Actual vs Predicted
Actual
LSTM Forecast
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
100
200
300
400
500

In [25]:
```
# RMSE Comparison Table
results = pd.DataFrame({
    'Model': ['ARIMA', 'XGBoost', 'LSTM'],
    'RMSE': [round(rmse_arima, 4), round(rmse_xgb, 4), round(rmse_lstm, 4)]
})
print(results.to_string(index=False))

# Bar chart
plt.figure(figsize=(8, 4))
colors = ['orange', 'green', 'red']
plt.bar(results['Model'], results['RMSE'], color=colors, width=0.4)
plt.title('Model RMSE Comparison')
plt.ylabel('RMSE (kW)')
plt.tight_layout()
plt.show()

# Overlay plot — Actual vs all 3 models (first 500 test points)
n = 500
actual_plot = y_test_lstm[:n]   # common reference — LSTM test set

plt.figure(figsize=(16, 5))
plt.plot(actual_plot, label='Actual', color='steelblue', linewidth=1.5)
plt.plot(y_pred_lstm[:n], label=f'LSTM (RMSE={rmse_lstm:.3f})', color='red', a
plt.plot(y_pred_xgb[-len(y_test_lstm):][:n], label=f'XGBoost (RMSE={rmse_xgb:.
plt.title('Actual vs Predicted — All Models')
plt.xlabel('Timesteps')
plt.ylabel('Global Active Power (kW)')
plt.legend()
plt.tight_layout()
plt.show()
```

```
  Model   RMSE
  ARIMA 0.9616
XGBoost 0.0340
   LSTM 0.5287
```




Actual vs Predicted — All Models
Actual
LSTM (RMSE=0.529)
XGBoost (RMSE=0.034)
Global Active Power (kW)
Timesteps
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
100
200
300
400
500

In [ ]:

In [ ]:

In [ ]: